universität wien
Faculty of Psychology

Lernen unter COVID-19-Bedingungen

Teachers

Method Report

1. Introduction

The present cross-sectional study is the second phase of the project "Learning under COVID-19 conditions" with pupils and students previously archived at https://doi.org/10.11587/VRQL3B and at https://doi.org/10.11587/XIU3TX, respectively.

The second phase of the project comprises three additional target groups – parents, teachers and school principals, in one cross-sectional measurement point. Thus, this phase of the project aims to investigate behaviour and coping of other agents in the school system, beside pupils', in order to get better picture in the situation and effects of COVID-19 on education system in Austria.

Study Description

1.1. Title and version

Lernen unter COVID-19-Bedingungen: LehrerInnen (Learning under COVID-19 conditions: teachers, V1.0)

1.2. Principal Investigators

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1.3. Funding/Acknowledgments

The present study has been funded by the Vienna Science and Technology Fund (WWTF), by MEGA Bildungsstiftung, and the City of Vienna through project COV20-025.

1.4. Fieldwork agency

The fieldwork of the study "Lernen unter COVID-19-Bedingungen" was managed and handled internally.

1.5. Topics/Keywords

- Sociodemographic information: age, sex, school type, region, gender role identity
- Teaching in crisis: overload, working conditions, working demands, self-efficacy, challenges
- School role: institutional trust, team resources
- Contact with parents: communicational patterns
- Digital teaching: digitalization, changes

2. Study Design

2.1. Sampling

The sample of the study "Lernen unter COVID-19-Bedingungen" consisted of teachers working in secondary education in Austria. The teachers were invited to participate in the survey by headmasters of their schools. Moreover, the link to and a recommendation to participate in the survey was published by the Federal Ministry of Education, Science, and Research on their website. The link with a respective invitation to the study was also placed on the survey's website. In order to participate in the study, the respondents had to agree to the informed consent, be Austrian residents, and be older than 18.

2.2. Fieldwork

The overall data collection was conducted in the time from 06 April 2021 to 25 May 2021.

3. Data Cleaning

3.1. General data cleaning procedure

Cleaning procedure was the same for all three data sets collected in three measurement points and contained two steps:

3.1.1. Initial data cleaning

As participating teachers answered an online questionnaire via Unipark (Questback, 2007), data was exported from unipark in .csv format and imported to SPSS. Variables names were replaced. All missing values were defined as -99. Invalid zero values or values of -77 were replaced with -99. Participants who gave consent and had answering time (variable **duration**) larger than zero were selected for a new SPSS file.

3.1.2. Data cleaning

As unipark assigns -1 to variable duration in cases where the break was made during the answering the questionnaire, all valid -1 cases were replaced with (+) 9999999 value. Participants with 100% missings on main outcome variables (related to self-efficacy) were excluded. Minimum possible duration was tested by filling in the questionnaire as fast as possible while reading all items. All cases above the duration treshold and above 18 years og age were selected for a new SPSS file.

3.1.3. Missing values

All empty spaces were replaced with -99. Missing values that were represented by 0 in the raw data set are replaced by -99 on all variables where 0 is not a valid value. Missing values represented with -77 in unipark were replaced with -99 as well. Missing values on variables that were result of filters applied, were defined with -98. Cases where

participants by mistake got to see a question they should not have seen (due to technical issues in unipark) were replaced with -78.

3.1.4. Filters

Data records of participants who did not give their consent are deleted.

A fastest possible time for answering the questionnaire is determined diligently. Variable duration of answers with duration below the fastest possible time is set to 9999999.

Data records with 100% missings on main outcome variables are deleted.

All other filters are defined in the questionnaire, as well as in the data set using values -98 and -78 (see above) as value labels.

3.1.5. Answers to open questions

All open questions have been removed from the data sets due to anonymization rules.

3.1.6. Multiple choice / single choice

Questions "In welcher/n Schulart/en unterrichten Sie?" (et.school.type1, et.school.type2, et.school.type3, et.school.type4, et.school.type5, et.school.type6, et.school.type7, et.school.type8, et.school.type9, et.school.type10), "In welchem/n Bundesland/ländern unterrichten Sie?" (et.school.fs1, et.school.fs2, et.school.fs3, et.school.fs4, et.school.fs5, et.school.fs6, et.school.fs7, et.school.fs8, et.school.fs9), "Sind Sie in der Deutschförderung tätig?" (et.ger.class1, et.ger.class2, et.ger.class3, et.ger.class4, et.ger.class5), and "Finden derzeit Elternabende statt?" (et.ptconf1, et.ptconf2, et.ptconf3, et.ptconf4, et.ptconf5, et.ptconf6) were posed in the multiple choice format.

3.2. Anonymisation procedure

3.2.1. Code variables

Due to the high risk for identification, variables et.code (personal code) and et.code.school (school code) were deleted from the data set.

3.2.2. Gender variables

In this dataset, answers indicating that the corresponding person chose "divers" (et.gender) are recoded to missing as there are fewer than 20 observations.

3.2.3. Age variables

Data containing the age of the participants in this dataset (**et.age**) are categorized by aggregating age cohorts to get groups with at least 20 observations.

3.2.4. School type variables

As there were fewer than 20 observations for "ausgewählt" category of the following school type variables (**et.school.type6** (Berufsschule und Lehre)), (**et.school.type7** (Polytechnische Schule (PTS)), (**et.school.type8** (Fachmittelschule (FMS)), (**et.school.type9** (Sonderschule)), they were removed from the dataset.

3.2.5. Variables related to German courses

As there were fewer than 20 observations for "ausgewählt" category of **et.ger.class3** (Deutschförderung: ja, Deutsch-förderklassen) and **et.ger.class5** (Deutschförderung: ja, andere), they were merged together as **et.ger.class5** (Deutschförderung: ja, Deutschförderklassen oder andere).

3.2.6. Federal state variables

As there were fewer than 20 observations for "ausgewählt" category of **et.school.fs1** (Burgenland), it was merged with **et.school.fs3** (Niederösterreich) to make (Burgenland oder Niederösterreich), and as there were fewer than 20 observations for "ausgewählt" category of **et.school.fs8** (Vorarlberg), it was merged with **et.school.fs7** (Tirol) to make (Tirol oder Vorarlberg).

3.2.7. Variable related to interaction with parents

As there were fewer than 20 observations for "ausgewählt" category of **et.ptconf4** (Finden derzeit Elternabende statt: ja, in Person/regulär), it was removed from the dataset.

3.2.8. Closed questions

The following answer categories were merged in this dataset to get groups with at least 20 observations:

Variable(s)	Answers merged	New value label
et.lost.pupils	• 0 to 25% • 26% to 50% • 51% to 75% • 76% to 100%	• 0-25% • 26-50% • 51-75% • 76-100%

3.2.9. Open questions

In this dataset 9 open variables are deleted without recoding or standardizing the answers:

- **et.school.type.other** (In welcher/n Schulart/en unterrichten Sie? Andere?)
- **et.ger.class.other** (Sind Sie in der Deutschförderung tätig? Ja, andere)

- et.ger.class.change (Wie hat sich die schulische Deutschförderung seit Beginn der Pandemie aus Ihrer Sicht verändert?)
- et.workh.change (Worauf ist diese Veränderung zurückzuführen?)
- et.cond3 (Falls etwas fehlt, was ist das aus Ihrer Sicht?)
- et.cond5 (Falls etwas fehlt, was ist das aus Ihrer Sicht?)
- et.poschange.other (Gibt es Dinge, die sich rückblickend im Laufe der COVID-19-Krise – auch, wenn sie vielleicht vorher schon gut waren - noch verbessert haben? Andere)
- ek.challenges (Was waren seit Beginn der Pandemie die größten Herausforderungen für Sie?)
- ek.accomplish (Was ist Ihnen seit Beginn der Pandemie besonders gut gelungen?)

3.2.10. Open questions containing numeric values

Answers to questions requiring numeric values are recoded according to the description in section "Data cleaning procedure". Original (string) answers are deleted.

3.3. Further deletions

3.3.1. Consent variable

One variable containing the participants' agreement on processing the filled in data for scientific purposes is deleted in each measurement point (et.consent) as all included participants gave their consent for processing the filled in data for scientific purposes.

3.3.2. Filter variables

One filter variable (filter_$) originating from the cleaning procedure is deleted in each measurement point.

3.3.3. Variables with few valid values

In this dataset, a variable that contains only missing values (quality) is deleted.

3.3.4. System variables

System variables that contain information of the duration of answering separate pages are deleted.

- external_lfdn (external sequence number)
- tester (filter variable indicating entries made for test purposes)
- dispcode (disposition code)
- lastpage (identifier of exit page)
- browser (browser identifier)
- referer (HTTP referer)
- page_history (array of page identifiers indicating the sequence of visited questionnaire pages)
- hflip (questionnaire answered on a screen in landscape mode)

- vflip (questionnaire answered on a screen in portrait mode)
- session_id (session identifier)
- Language (language)
- ats (absolute timestamp)
- datetime (date of first access)
- date_of_last_access (date of last access)
- date_of_first_mail (date of first e-mail)
- rts4883664, rts4883668, rts4883677, rts4883687, rts4883768, rts4883956, rts4884182, rts4884235, rts4884255, rts4884289, rts4906700, rts4907304, rts4907351, rts4907375, rts4907437, rts4907454, rts4907593, rts4959567, rts4985562, rts4985692, rts4985699, rts5034875 (relative timestamps for pages)

3.3.5. Reverse-coded variables

All reverse-coded variables are deleted.

- et.load.sp1.rec (Reversed Likert scale of item et.load.sp1)
- et.load.sp2.rec (Reversed Likert scale of item et.load.sp2)
- et.load.sp3.rec (Reversed Likert scale of item et.load.sp3)
- et.load.sp4.rec (Reversed Likert scale of item et.load.sp4)
- et.load.sp5.rec (Reversed Likert scale of item et.load.sp5)
- et.load.sp6.rec (Reversed Likert scale of item et.load.sp6)
- et.load.sp7.rec (Reversed Likert scale of item et.load.sp7)
- et.load.sp8.rec (Reversed Likert scale of item et.load.sp8)
- et.load.sp9.rec (Reversed Likert scale of item et.load.sp9)
- et.load.sp10.rec (Reversed Likert scale of item et.load.sp10)
- et.load.sp11.rec (Reversed Likert scale of item et.load.sp11)
- et.load.sp12.rec (Reversed Likert scale of item et.load.sp12)
- et.load.sp13.rec (Reversed Likert scale of item et.load.sp13)
- et.load.sp14.rec (Reversed Likert scale of item et.load.sp14)
- et.workh.rec (Reversed Likert scale of item et.workh)
- et.cond1.rec (Reversed Likert scale of item et.cond1)
- et.cond2.rec (Reversed Likert scale of item et.cond2)
- et.cond4.rec (Reversed Likert scale of item et.cond4)
- et.eff1.rec (Reversed Likert scale of item et.eff1)
- et.eff2.rec (Reversed Likert scale of item et.eff2)
- et.eff3.rec (Reversed Likert scale of item et.eff3)
- et.eff4.rec (Reversed Likert scale of item et.eff4)
- et.school.safety1.rec (Reversed Likert scale of item et.school.safety1)
- et.school.safety2.rec (Reversed Likert scale of item et.school.safety2)

- et.school.safety3.rec (Reversed Likert scale of item et.school.safety3)
- et.school.safety4.rec (Reversed Likert scale of item et.school.safety4)
- et.school.safety5.rec (Reversed Likert scale of item et.school.safety5)
- et.school.info1.rec (Reversed Likert scale of item et.school.info1)
- et.school.info2.rec (Reversed Likert scale of item et.school.info2)
- et.psysaf1.rec (Reversed Likert scale of item et.psysaf1)
- et.psysaf2.rec (Reversed Likert scale of item et.psysaf2)
- et.psysaf3.rec (Reversed Likert scale of item et.psysaf3)
- et.psysaf4.rec (Reversed Likert scale of item et.psysaf4)
- et.psysaf5.rec (Reversed Likert scale of item et.psysaf5)
- et.teamlearn1.rec (Reversed Likert scale of item et.teamlearn1)
- et.teamlearn2.rec (Reversed Likert scale of item et.teamlearn2)
- et.teamlearn3.rec (Reversed Likert scale of item et.teamlearn3)
- et.teamlearn4.rec (Reversed Likert scale of item et.teamlearn4)
- et.teamerror1.rec (Reversed Likert scale of item et.teamerror1)
- et.teamerror2.rec (Reversed Likert scale of item et.teamerror2)
- et.teamerror3.rec (Reversed Likert scale of item et.teamerror3)
- et.teamerror4.rec (Reversed Likert scale of item et.teamerror4)
- et.parwork1.rec (Reversed Likert scale of item et.parwork1)
- et.parwork2.rec (Reversed Likert scale of item et.parwork2)
- et.parwork3.rec (Reversed Likert scale of item et.parwork3)
- et.parwork4.rec (Reversed Likert scale of item et.parwork4)
- et.parwork5.rec (Reversed Likert scale of item et.parwork5)
- et.ptrealtion1.rec (Reversed Likert scale of item et.ptrealtion1)
- et.ptrealtion2.rec (Reversed Likert scale of item et.ptrealtion2)
- et.ptrealtion3.rec (Reversed Likert scale of item et.ptrealtion3)
- et.ptrealtion4.rec (Reversed Likert scale of item et.ptrealtion4)
- et.ptrealtion5.rec (Reversed Likert scale of item et.ptrealtion5)
- et.teamwork1.rec (Reversed Likert scale of item et.teamwork1)
- et.teamwork2.rec (Reversed Likert scale of item et.teamwork2)
- et.teamwork3.rec (Reversed Likert scale of item et.teamwork3)
- et.teamwork4.rec (Reversed Likert scale of item et.teamwork4)
- et.teamwork5.rec (Reversed Likert scale of item et.teamwork5)
- et.pupneeds1.rec (Reversed Likert scale of item et.pupneeds1)
- et.pupneeds2.rec (Reversed Likert scale of item et.pupneeds2)
- et.pupneeds3.rec (Reversed Likert scale of item et.pupneeds3)
- et.pupneeds4.rec (Reversed Likert scale of item et.pupneeds4)
- et.pupneeds5.rec (Reversed Likert scale of item et.pupneeds5)

- et.appreciation1.rec (Reversed Likert scale of item et.appreciation1)
- et.appreciation2.rec (Reversed Likert scale of item et.appreciation2)
- et.ptconf6.rec (Reversed Likert scale of item et.ptconf6)
- et.digipre1.rec (Reversed Likert scale of item et.digipre1)
- et.digipre2.rec (Reversed Likert scale of item et.digipre2)
- et.digipre3.rec (Reversed Likert scale of item et.digipre3)
- et.digipost1.rec (Reversed Likert scale of item et.digipost1)
- et.digipost2.rec (Reversed Likert scale of item et.digipost2)
- et.digichange1.rec (Reversed Likert scale of item et.digichange1)
- et.digichange2.rec (Reversed Likert scale of item et.digichange2)
- et.digichange3.rec (Reversed Likert scale of item et.digichange3)
- et.poschange1.rec (Reversed Likert scale of item et.poschange1)
- et.poschange2.rec (Reversed Likert scale of item et.poschange2)
- et.poschange3.rec (Reversed Likert scale of item et.poschange3)
- et.poschange4.rec (Reversed Likert scale of item et.poschange4)
- et.poschange5.rec (Reversed Likert scale of item et.poschange5)
- et.poschange6.rec (Reversed Likert scale of item et.poschange6)
- et.poschange7.rec (Reversed Likert scale of item et.poschange7)
- et.optimism1.rec (Reversed Likert scale of item et.optimism1)
- et.optimism2.rec (Reversed Likert scale of item et.optimism2)
- et.optimism3.rec (Reversed Likert scale of item et.optimism3)
- et.optimism4.rec (Reversed Likert scale of item et.optimism4)
- et.optimism5.rec (Reversed Likert scale of item et.optimism5)
- et.optimism6.rec (Reversed Likert scale of item et.optimism6)
- et.optimism7.rec (Reversed Likert scale of item et.optimism7)
- et.wol1.rec (Reversed Likert scale of item et.wol1)
- et.wol2.rec (Reversed Likert scale of item et.wol2)
- et.wol3.rec (Reversed Likert scale of item et.wol3)
- et.wol4.rec (Reversed Likert scale of item et.wol4)
- et.wol5.rec (Reversed Likert scale of item et.wol5)
- et.m1.rec (Reversed Likert scale of item et.m1)
- et.m2.rec (Reversed Likert scale of item et.m2)
- et.m3.rec (Reversed Likert scale of item et.m3)
- et.m4.rec (Reversed Likert scale of item et.m4)
- et.m5.rec (Reversed Likert scale of item et.m5)
- et.m6.rec (Reversed Likert scale of item et.m6)
- et.f1.rec (Reversed Likert scale of item et.f1)
- et.f2.rec (Reversed Likert scale of item et.f2)

- et.f3.rec (Reversed Likert scale of item et.f3)
- et.f4.rec (Reversed Likert scale of item et.f4)
- et.f5.rec (Reversed Likert scale of item et.f5)
- et.f6.rec (Reversed Likert scale of item et.f6)

3.3.6. Mean scale variables

All variables that contain calculated mean values for each scale are deleted.

- et.wol (Mean scale Arbeitsaufwand (reversed items used))
- et.ag (Mean scale Agency (reversed items used))
- et.co (Mean scale Communion (reversed items used))
- et.cond (Mean scale Zufriedenheit Rahmenbedingungen (reversed items used))
- et.eff (Mean scale Handlungsfähigkeit (reversed items used))
- et.school.safety (Mean scale School safety (reversed items used))
- et.school.info (Mean scale School info (reversed items used))
- et.teamlearn (Mean scale Arbeitsprozesse (reversed items used))
- et.teamerror (Mean scale Umgang mit Fehlern (reversed items used))
- et.digipre (Mean scale Digitalisierung prä (reversed items used))
- et.digipost (Mean scale Digitalisierung post (reversed items used))
- et.digichange (Mean scale Digitalisierung change (reversed items used))
- et.poschange (Mean scale Positive change (reversed items used))
- et.optimism (Mean scale Optimism (reversed items used))

3.4. Increasing reusability

To increase the reusability of the data set variables are rearranged into an appropriate order. Variable names are renamed to get more meaningful names.